Table S5. Representation diagnostics: step-5 checks (protocol §5) and, separately, the exploratory X1 eraser on the 158-case analysis region.

Check	Set (cases)	FM-P	FM-S	REF	FM-P-nat
Step-5 checks (protocol §5)					
E1: site BA, linear probe[b]	278, all steps	0.960, pass (0.932 to 0.984)	0.768, pass (0.713 to 0.819)	0.868, pass (0.830 to 0.903)	0.993, pass (0.982 to 1.000)
E1: site BA, linear probe[b]	158, analysis region	0.914, pass (0.864 to 0.958)	0.821, pass (0.757 to 0.876)	0.837, pass (0.781 to 0.893)	0.966, pass (0.930 to 0.993)
E1: site BA, linear probe[b]	123, single-file[a]	0.881, pass (0.806 to 0.949)	0.632, pass (0.547 to 0.722)	0.838, pass (0.755 to 0.911)	0.981, pass (0.942 to 1.000)
E2: ΔAUC before removal[c]	158, analysis region	0.050, met (0.019 to 0.080)	0.002, not met (−0.017 to 0.024)	0.024, not met (−0.000 to 0.049)	0.044, not met (0.013 to 0.077)
E2: ΔAUC after protocol removal[c]	158, analysis region	0.044, not met (0.012 to 0.074)	0.003, not met (−0.018 to 0.025)	0.024, not met (−0.000 to 0.051)	—
E2: ΔAUC before removal[c]	123, single-file[a]	0.041, not met (0.011 to 0.071)	−0.004, not met (−0.022 to 0.017)	0.010, not met (−0.014 to 0.036)	0.041, not met (0.009 to 0.072)
E2: ΔAUC after protocol removal[c]	123, single-file[a]	0.025, not met (−0.005 to 0.056)	−0.005, not met (−0.022 to 0.014)	0.010, not met (−0.015 to 0.036)	—
E3-site: site BA after removal, linear probe	278, all steps	0.250 (0.250 to 0.250)	0.250 (0.250 to 0.250)	0.250 (0.250 to 0.250)	—
E3-site: one-sided 95% upper bound, linear[d]	278, all steps	0.250, pass	0.250, pass	0.250, pass	—
E3-site: site BA after removal, linear probe	158, analysis region	0.416 (0.352 to 0.481)	0.393 (0.346 to 0.444)	0.448 (0.387 to 0.518)	—
E3-site: one-sided 95% upper bound, linear[d]	158, analysis region	0.472, fail	0.435, fail	0.505, fail	—
E3-site: site BA after removal, linear probe	123, single-file[a]	0.380 (0.315 to 0.451)	0.323 (0.259 to 0.395)	0.405 (0.329 to 0.488)	—
E3-site: one-sided 95% upper bound, linear[d]	123, single-file[a]	0.441, fail	0.385, fail	0.473, fail	—
E3-site: site BA after removal, MLP probe	278, all steps	0.957 (0.927 to 0.982)	0.495 (0.456 to 0.535)	0.863 (0.818 to 0.904)	—
E3-site: one-sided 95% upper bound, MLP[d]	278, all steps	0.979, > 0.35	0.531, > 0.35	0.900, > 0.35	—

Check	Set (cases)	FM-P	FM-S	REF	FM-P-nat
E3-site: site BA after removal, MLP probe	158, analysis region	0.702 (0.640 to 0.763)	0.428 (0.368 to 0.489)	0.620 (0.543 to 0.687)	—
E3-site: one-sided 95% upper bound, MLP[d]	158, analysis region	0.755, > 0.35	0.476, > 0.35	0.680, > 0.35	—
E3-site: site BA after removal, MLP probe	123, single-file[a]	0.569 (0.510 to 0.631)	0.446 (0.376 to 0.522)	0.542 (0.458 to 0.628)	—
E3-site: one-sided 95% upper bound, MLP[d]	123, single-file[a]	0.621, > 0.35	0.511, > 0.35	0.615, > 0.35	—
E3-state: retained fraction of ΔAUC[f]	158, analysis region	0.868, pass	not applicable[e]	not applicable[e]	—
E3-state: retained fraction of ΔAUC[f]	123, single-file[a]	not applicable[e]	not applicable[e]	not applicable[e]	—
Reported only (no criterion)					
E1: site BA, MLP probe (before removal)	278, all steps	0.987 (0.970 to 1.000)	0.846 (0.799 to 0.889)	0.933 (0.901 to 0.961)	0.978 (0.957 to 0.995)
E1: site BA, MLP probe (before removal)	158, analysis region	0.885 (0.829 to 0.936)	0.853 (0.792 to 0.909)	0.894 (0.847 to 0.938)	0.959 (0.924 to 0.987)
E1: site BA, MLP probe (before removal)	123, single-file[a]	0.877 (0.801 to 0.945)	0.663 (0.594 to 0.735)	0.817 (0.738 to 0.892)	0.962 (0.904 to 1.000)
E2: ΔAUC, all steps	278, all steps	0.035 (0.018 to 0.050)	0.006 (−0.002 to 0.016)	0.010 (−0.001 to 0.022)	0.034 (0.016 to 0.052)
E2: ΔAUC, normalized-time base[g]	278, all steps	0.018 (0.010 to 0.027)	0.004 (−0.001 to 0.009)	0.006 (−0.001 to 0.013)	0.018 (0.008 to 0.027)
E2: ΔAUC, normalized-time base[g]	158, analysis region	0.058 (0.027 to 0.085)	0.014 (−0.006 to 0.032)	0.031 (0.007 to 0.055)	0.050 (0.017 to 0.083)
E2: ΔAUC, normalized-time base[g]	123, single-file[a]	0.045 (0.011 to 0.080)	0.006 (−0.018 to 0.030)	0.023 (−0.007 to 0.051)	0.044 (0.008 to 0.080)
E2: ΔAUC, base with past actions[h]	158, analysis region	−0.081 (−0.146 to −0.016)	−0.059 (−0.118 to −0.004)	−0.048 (−0.108 to 0.013)	−0.074 (−0.141 to −0.005)
Exploratory X1 (outside the protocol; eraser refitted on the 158-case analysis region)[i]					
X1 E3-site: site BA after removal, linear probe	158, analysis region	0.250 (0.250 to 0.250)	0.250 (0.250 to 0.250)	0.250 (0.250 to 0.250)	—
X1 E3-site: one-sided 95% upper bound, linear	158, analysis region	0.250	0.250	0.250	—

Check	Set (cases)	FM-P	FM-S	REF	FM-P-nat
X1 E3-site: site BA after removal, MLP probe	158, analysis region	0.577 (0.506 to 0.647)	0.309 (0.255 to 0.366)	0.458 (0.392 to 0.530)	—
X1 E3-site: one-sided 95% upper bound, MLP	158, analysis region	0.637, > 0.35	0.357, > 0.35	0.517, > 0.35	—
X1 E2: ΔAUC after X1 removal[c]	158, analysis region	0.051, met (0.019 to 0.083)	−0.000, not met (−0.022 to 0.021)	0.024, not met (−0.002 to 0.052)	—
X1 E3-state: retained fraction of ΔAUC[f]	158, analysis region	1.019, pass	not applicable[e]	not applicable[e]	—

Values: estimate (95% interval); balanced accuracy (BA) of the case-level four-centre site probe (chance 0.25); intervals are case-level bootstrap intervals stratified by site (2,000 resamples, seed 20261008; percentile 2.5–97.5). Step-5 values from the step-5 record; X1 values from the X1/X2 record. FM-P-nat has no site-removed arm (—).

[a]§10.3 single-file sensitivity set (123 cases of the primary set).

[b]E1 criterion: BA ≥ 0.50 with lower bound > 0.35.

[c]ΔAUC = held-out AUC for S_t of base + R_t minus base (elapsed-time spline); criterion ΔAUC ≥ 0.05 with the interval excluding 0.

[d]E3-site criterion (linear probe): one-sided 95% upper bound (95th percentile of the bootstrap) ≤ 0.35 on both the 278-case and the 158-case set; the MLP probe is compared with 0.35 for the claim ceiling only.

[e]E2 before removal not met, so retention of state information could not be assessed.

[f]ΔAUC after removal / ΔAUC before removal; criterion ≥ 0.50.

[g]Normalized time (step / case length) uses future information and is not a valid baseline (protocol §5.2).

[h]Base = elapsed time + past synthetic actions in one reference cell (K = 5, P = 0, W = 5, α = 0.5, γ = 1, φ_A = 0, replicate 1).

[i]Exploratory (plan registered on OSF, osf.io/5c627): the X1 eraser is fitted on training folds and evaluated on held-out folds of the same population; its linear E3-site value is a check of the procedure and does not gate any comparison. No verdict.

Abbreviations: AUC, area under the receiver operating characteristic curve; MLP, two-layer perceptron probe; FM-P, FM-S, REF, primary, secondary and reference representations; nat, native acquisition.